

Internship Plan

Course code and name:

814601S Work Experience in ICT responsibilities (5 ECTS)

814311A Internship in ICT duties (5 ECTS)

Student name: Arslan Sultan

Student number: 1.2.246.562.24.35253236476

Degree programme: Information Processing Science / Computer Science and Engineering

University: University of Oulu

Student email: arslan.sultan@student.oulu.fi

Type of internship: University-supported summer trainee scholarship (financial support from the University of Oulu)

Employer / host organization: Universidad Politécnica de Madrid (UPM), Escuela Técnica Superior de Ingeniería de Sistemas Informáticos (ETSISI)

Workplace address: Calle Alan Turing S/N, 28040 Madrid, Spain

Workplace supervisor: Juan Garbajosa, Professor

University contact / teacher responsible: University Lecturer Tonja Molin-Juustila (tonja.molin-juustila@oulu.fi)

1. Internship period

Item	Details
Start date	1 May 2026
End date	31 July 2026
Total duration	13 weeks
Working time	Full-time, 40 hours per week
Estimated total hours	520 hours

2. Background and purpose

The purpose of this internship is to apply knowledge and skills acquired during university studies to ICT work assignments in software development and research software engineering. The main project is **The Simulation of Collective Intelligence in Agile Teams: a Transparent Testbed**, an academic

prototype that models collective intelligence (CI), agile team effectiveness, and human–AI teaming across software development sprints.

The internship connects directly to my degree studies and thesis work. It provides practical experience in Python development, simulation design, technical documentation, literature-based modelling, and academic dissemination.

3. Learning objectives

The internship is planned to fulfil the following learning objectives for practical training:

1. **Apply university knowledge to field work** — use software engineering, research methods, data analysis, and agile development concepts in a real ICT assignment.
 2. **Evaluate and develop myself as a learner and worker** — reflect on feedback, improve work methods, and balance implementation with research quality.
 3. **Plan and evaluate use of time and working methods** — follow weekly goals, maintain progress reports, and manage a multi-month development project.
 4. **Work systematically and goal-oriented, alone and in collaboration** — implement software modules independently and discuss model assumptions with supervisor and research contacts.
 5. **Understand factors directing the work community** — follow employer priorities, reporting requirements, confidentiality rules, and academic quality standards.
 6. **Identify possible post-graduation duties** — explore research software engineering, simulation/HCI, and agile/team-effectiveness roles.
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4. Planned duties and assignments

4.1 Software development

- Design and implement **The Simulation of Collective Intelligence in Agile Teams: a Transparent Testbed** in Python with a Streamlit user interface.
- Build core modules: simulation engine, team/backlog generation, AI support, metrics, and experiments.
- Add advanced features: externalized weights, preset scenarios, mixed backlogs, task dependencies, defect rework, Scrum roles, and export functions.
- Implement automated tests and run validation experiments (Monte Carlo, sensitivity analysis).

4.2 Research modelling and documentation

- Review collective intelligence and agile effectiveness literature.
- Map research constructs to testbed variables and document assumptions transparently.

- Produce UML/conceptual diagrams and performance analysis documents.
- Integrate supervisor feedback aligning the model with Riedl (2021), Kommol et al. (2025), Hackman/Wageman, and agile/Scrum literature.

4.3 Academic dissemination support

- Prepare demo/poster submission materials for a collective intelligence conference context.
- Create parameter dictionaries, architecture descriptions, and presentation-ready summaries.

4.4 Deliverables

Deliverable	Description
Transparent testbed (model v2.0)	Python/Streamlit prototype with tests and externalized weights
Documentation	UML, CI analysis, parameter dictionary, presentation report
CI/HCOMP conference submissions	Two submissions: Demo track + Poster track (<i>Collective Intelligence in Agile Teams: A Transparent Simulation Testbed for Studying Team Decisions Across Sprints</i>)
Internship reports	Plan, 13 weekly reports, final report

5. Planned weekly schedule

Week	Period	Main focus
1	1–7 May 2026	Onboarding, environment setup, project scoping, literature review start
2	8–14 May 2026	Architecture review, baseline codebase structure, initial simulation loop
3	15–21 May 2026	Core modules: team, tasks, metrics, AI support
4	22–28 May 2026	Research paper analysis, presentation report, supervisor feedback review
5	29 May – 4 Jun 2026	Process criteria, team engagement, CI predictor integration
6	5–11 Jun 2026	New references, Riedl/Kommol/Hackman model alignment
7	12–18 Jun 2026	Consequentiality, age diversity, terminology and formula refinements
8	19–25 Jun 2026	UML and conceptual documentation
9	26 Jun – 2 Jul 2026	CI performance analysis document, PDF export pipeline
10	3–9 Jul 2026	Model v2.0: externalized weights, presets, UI charts, exports
11	10–16 Jul 2026	Mixed backlog, dependencies, rework, pytest suite, sensitivity
12	17–23 Jul 2026	Demo submission materials, improvement roadmap
13	24–31 Jul 2026	Final validation, documentation polish, final report preparation

6. Reporting plan

Because this internship is supported by the University of Oulu, reporting follows the **plan + weekly progress reports + final report** model.

Deliverable	When
Internship plan (this document)	Before start / week 1
Weekly progress reports	One report per week; submitted the Monday after each reporting week
Final internship report	After 31 July 2026
Employer testimonial	At end of internship
Transcript of study records	Attached to final report

Each weekly progress report will include:

- reporting period dates;

- hours worked (35–40 h per week);
 - summary of completed work;
 - connection to learning objectives;
 - challenges and solutions;
 - plan for the next week.
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7. Required tools and resources

- Python 3.11+, Streamlit, pandas, PyYAML, pytest
 - Git version control
 - Literature references (`new-ref.bib`, research papers)
 - University Moodle internship learning environment for submission
 - Workplace computer / development environment
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8. Risk management

Risk	Mitigation
Scope grows beyond available time	Prioritize core testbed and documentation; defer optional extensions
Theory-to-code translation is unclear	Document assumptions explicitly; discuss with supervisor before major model changes
Environment/dependency issues	Use virtual environments; validate in isolated test runs
Confidentiality concerns	Exclude sensitive employer information from public reports; ask permission when uncertain